

Greenberg Harper 8

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1 Affine Preliminaries

Let $\mathbb{R}^n$ act transitively and faithfully on a set E . E is said to be affine space. Let elements of E be called points. Suppose $P, Q \in E$. Define $Q - P$ to be the element $v \in \mathbb{R}^n$ such that $v + P = Q$. For $P \neq Q$, let line PQ denote $\{t(Q - P) + P : t \in \mathbb{R}\}$.

Let E, E' be affine spaces. An affine map $f : E \rightarrow E'$ is a function such that for all $t \in \mathbb{R}$, $P, Q \in E$, that $f((1 - t)P + tQ) = (1 - t)f(P) + tf(Q)$.

Fix $0 \in E$ and $0' \in E'$. Let $\mathcal{A} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear operator. Define $f : E \rightarrow E'$ by $f(P) = \mathcal{A}(P - 0) + 0'$. Then f is an affine map.

Proof:

$$\begin{aligned} f((1 - t)P + tQ) &= f(t(Q - P) + P) = \mathcal{A}((t(Q - P) + P) - 0) + 0' \\ &= \mathcal{A}(t(Q - P) + (P - 0)) + 0' = t\mathcal{A}(Q - P) + \mathcal{A}(P - 0) + 0' \\ &= t\mathcal{A}(Q - P) + f(P) = t\mathcal{A}((Q - 0) - (P - 0)) + f(P) \\ &= t((\mathcal{A}(Q - 0) + 0') - (\mathcal{A}(P - 0) + 0')) + f(P) \\ &= t(f(Q) - f(P)) + f(P) = (1 - t)f(P) + tf(Q). \end{aligned}$$

Conversely, suppose f is an affine map. Fix $0 \in E$ and let $0' = f(0)$. Let $\mathcal{A} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be given by $\mathcal{A}v = f(v + 0) - 0'$.

Suppose $\lambda \in \mathbb{R}$. Then

$$\begin{aligned} f(\lambda(v + 0) - 0) &= \lambda(f(v + 0) - f(0)) + f(0) \\ f(\lambda v + 0) &= \lambda\mathcal{A}v + f(0) \\ f(\lambda v + 0) - f(0) &= \lambda\mathcal{A}v \\ \mathcal{A}\lambda v &= \lambda\mathcal{A}v. \end{aligned}$$

Finally,

$$\mathcal{A}(v + w) = f((v + w) + 0) - f(0) =$$

2 Singular Homology Theory

Define the module homomorphism $\partial^\# : S_0(X) \rightarrow R$ by $x \mapsto 1$ for a simplicial 0-simplex x . Show that $\partial^\# \partial = 0$. Let $Z_0^\#(X) = \ker \partial^\#$. Then define $H_0^\#(X) = Z_0^\#(X)/B_0(X)$. Define $H_q^\#(X) = H_q(X)$ for $q > 1$. Suppose X has n path components. If $n = 1$, then $H_0^\#(X) \cong 0$. If $n > 1$, then $H_0^\#(X) \cong R^{n-1}$.

Proof: Let $\gamma : I \rightarrow X$ be a simplicial 1-simplex, ie a path, say going from x_0 to x_1 . Then $\partial^\# \partial \gamma = \partial^\#(x_1 - x_0) = 1 - 1 = 0$. By linearity $\partial^\# \partial = 0$ for all $c \in S_1(X)$.

Suppose $n = 1$. Observe that in general $B_0(X)$ is generated by elements of the form $\{x_f - x_i\}$ if x_i, x_f are path connected. Let $c \in Z_0^\#(X)$. Fix $x_0 \in X$. Since X is path connected, in $H_0^\#(X)$ we have

$$c = \sum r_i x_i = \sum r_i x_0 = \left(\sum r_i \right) x_0 = 0,$$

since $\sum r_i = 0$ by assumption, so $H_0^\#(X) = 0$.

Suppose $n > 1$. Since $\partial^\# \partial = 0$, $\partial^\#$ descends to a map $H_0^\#(X) \rightarrow R$. The previous result gives $H_0^\#(X) \cong R^n$. Fix the standard basis $\{e_i\}$ of R^n . Then $e_i \mapsto x_i \mapsto 1$